SIMATS ENGINEERING

CO- 2: AT – 2: Simulation Task - Medium

COURSE NAME: Advance Wireless Communication Systems /

5G / 6G Communication Systems

COURSE CODE: ECA56

COURSE OUTCOME COVERED

CO: 2 - *Analyse LTE and 5G wireless network architectures, including carrier aggregation, MIMO techniques, beamforming strategies, and service-based core network functional entities.* (BL4)

Assessment Tool Weightage: 35%

Each student should simulate one question and should upload the results in Github account along with the necessary outputs and codes. This assessment tool is evaluated for 50 marks.

QUESTIONS: **Total Marks: 50** **Duration: 90 Mins**

1. Simulation of LTE throughput performance under varying bandwidth, SNR, modulation order, coding rate, and user load (BL4)
2. Simulation of LTE protocol stack delay analysis under uplink traffic, downlink traffic, packet size, retransmission count, and channel quality (BL4)
3. Simulation of carrier aggregation performance in LTE-Advanced under single carrier, dual carrier, triple carrier, varying bandwidth, and SNR (BL4)
4. Simulation of LTE resource block allocation under different user numbers, scheduling methods, bandwidths, traffic load, and interference levels (BL4)
5. Simulation of LTE scheduler performance using Round Robin, Proportional Fair, Max C/I, delay variation, and throughput comparison (BL4)
6. Simulation of BER performance in 2×2, 4×4, 8×8 MIMO under varying SNR, modulation schemes, fading conditions, and antenna correlation (BL4)
7. Simulation of MIMO channel capacity under antenna number, SNR, bandwidth, fading model, and multiplexing order (BL4)
8. Simulation of SU-MIMO performance under channel gain, BER, throughput, latency, and SNR (BL4)
9. Simulation of MU-MIMO throughput comparison under user count, antenna count, SNR, modulation, and interference (BL4)
10. Simulation of higher-order MIMO diversity gain under fading channel, BER, outage probability, SNR, and antenna configuration (BL4)

11. Simulation of beamforming gain under steering angle, antenna elements, frequency, SNR, and interference (BL4)
12. Simulation of analog beamforming performance under phase shift variation, beam angle, antenna count, gain, and sidelobe level (BL4)
13. Simulation of digital beamforming under user direction, beam width, SINR, antenna spacing, and channel condition (BL4)
14. Simulation of hybrid beamforming under RF chains, beam direction, power allocation, SNR, and throughput (BL4)
15. Simulation of beam steering in antenna arrays under scan angle, frequency, gain, beam width, and sidelobe suppression (BL4)
16. Simulation of 5G throughput performance under QPSK, 16-QAM, 64-QAM, 256-QAM, and varying SNR (BL4)
17. Simulation of 5G NR air interface performance under different numerologies, bandwidths, latency, subcarrier spacing, and throughput (BL4)
18. Simulation of subcarrier spacing effect under 15 kHz, 30 kHz, 60 kHz, 120 kHz, and symbol duration (BL4)
19. Simulation of OFDMA resource allocation under user density, subcarriers, power allocation, throughput, and delay (BL4)
20. Simulation of waveform comparison between LTE OFDM and 5G NR under PAPR, BER, bandwidth, latency, and spectral efficiency(BL4)
21. Simulation of gNodeB traffic handling under user density, latency, throughput, packet loss, and scheduling (BL4)
22. Simulation of functional split in 5G RAN under processing delay, fronthaul load, throughput, latency, and energy consumption (BL4)
23. Simulation of RRC procedure timing under handover delay, signaling load, connection setup time, throughput, and failure rate (BL4)
24. Simulation of service-based architecture performance under AMF load, SMF load, UPF throughput, latency, and session count (BL4)
25. Simulation of network slicing under bandwidth allocation, latency, user priority, throughput, and resource utilization (BL4)
26. Simulation of AMF, SMF, and UPF traffic load under session count, packet delay, throughput, CPU utilization, and latency (BL4)
27. Simulation of control plane and user plane separation under packet delay, signaling load, throughput, latency, and efficiency (BL4)
28. Simulation of 5G core traffic routing under user density, mobility, latency, throughput, and resource utilization (BL4)
29. Simulation of packet delivery performance in 5G core under load variation, routing delay, throughput, packet loss, and queue size (BL4)
30. Simulation of mobility management under handover count, delay, throughput, signaling load, and failure probability (BL4)
31. Simulation of propagation loss under 700 MHz, 3.5 GHz, 28 GHz, 38 GHz, and 60 GHz (BL4)
32. Simulation of mmWave path loss under distance, frequency, atmospheric absorption, blockage, and antenna gain (BL4)
33. Simulation of channel modeling for mmWave MIMO under LOS, NLOS, fading, delay spread, and SNR (BL4)
34. Simulation of interference management under user density, reuse factor, SNR, SINR, and throughput (BL4)
35. Simulation of fading channel effects under Rayleigh, Rician, Nakagami, BER, and outage probability(BL4)

36. Simulation of V2X communication under latency, vehicle speed, packet delivery ratio, throughput, and interference (BL4)
37. Simulation of NB-IoT performance under coverage, delay, power consumption, throughput, and packet loss (BL4)
38. Simulation of edge computing delay under local processing, cloud processing, latency, bandwidth, and task size (BL4)
39. Simulation of smart city IoT traffic under sensor count, delay, packet delivery, bandwidth, and throughput (BL4)
40. Simulation of AI-based wireless optimization under prediction accuracy, throughput, delay, resource allocation, and energy efficiency (BL4)

Code of Conduct:

I B. Yagneswar Reddy(192312285) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *Yagneswar reddy*

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding Model Design	20%	Demonstrates strong understanding of underlying concepts in the simulation 5	Shows good understanding with minor gaps 4	Partial understanding; some misconceptions 3	Lacks understanding of key concepts 1-2
Tool Usage	20%	Uses simulation tools effectively with correct setup and execution 5	Uses tools with minor errors or guidance 4	Limited ability to use tools properly 3	Unable to use simulation tools 1-2
Analysis of Results	20%	Provides clear, logical analysis and meaningful interpretation of results 5	Analysis is reasonable with minor gaps 4	Superficial analysis; limited interpretation 3	Unable to analyze or interpret results 1-2
Documentation	20%	Presents results clearly with proper documentation 5	Documentation is adequate with minor issues 4	Poorly organized or incomplete documentation 3	No proper documentation or presentation 1-2

Criteria	**Question Number :**			
	Level 4	**Level 3**	**Level 2**	**Level 1**
Conceptual Understanding Model Design				
Tool Usage				
Analysis of Results				
Documentation				

Code:

```
clc;
clear;
close all;

%% =============================================================
%  BER Performance Simulation of 2x2, 4x4 and 8x8 MIMO Systems
%  Varying SNR, Modulation, Fading and Antenna Correlation
% ==============================================================

%% Simulation Parameters
Nt_values = [2 4 8];          % Number of Transmit Antennas
Nr_values = [2 4 8];          % Number of Receive Antennas
SNR_dB = 0:2:20;              % SNR range in dB

modulations = {'BPSK','QPSK','16QAM'};
fading_types = {'Rayleigh','Rician'};
correlation_values = [0 0.5 0.9];

numBits = 1e5;                % Number of bits per simulation

%% =============================================================
%  Main Simulation
% ==============================================================

for modIndex = 1:length(modulations)

   modulation = modulations{modIndex};

   % Determine bits per symbol
   switch modulation
      case 'BPSK'
         bitsPerSymbol = 1;
      case 'QPSK'
         bitsPerSymbol = 2;
      case '16QAM'
         bitsPerSymbol = 4;
   end

   fprintf('\n=====================================\n');
   fprintf('Modulation: %s\n', modulation);
   fprintf('=====================================\n');
```

```
for fadeIndex = 1:length(fading_types)

    fading = fading_types{fadeIndex};

    for corrIndex = 1:length(correlation_values)

        rho = correlation_values(corrIndex);

        figure;
        hold on;
        grid on;

        legendText = {};

        for antennaIndex = 1:length(Nt_values)

            Nt = Nt_values(antennaIndex);
            Nr = Nr_values(antennaIndex);

            BER = zeros(1,length(SNR_dB));

            %% Generate random bits
            bits = randi([0 1], numBits, 1);

            %% Modulation
            txSymbols = modulateSignal(bits, modulation);

            % Reshape symbols into Nt transmit streams
            numSymbols = floor(length(txSymbols)/Nt)*Nt;
            txSymbols = txSymbols(1:numSymbols);

            txMatrix = reshape(txSymbols, Nt, []);

            %% SNR Loop
            for snrIndex = 1:length(SNR_dB)

                snrLinear = 10^(SNR_dB(snrIndex)/10);

                %% Generate MIMO Channel
                H = generateChannel(Nr, Nt, fading, rho);

                %% Generate received signal
                noise = (randn(Nr,size(txMatrix,2)) + ...
                         1i*randn(Nr,size(txMatrix,2))) / sqrt(2);
```

```
    % Normalize channel
    H = H / sqrt(Nt);

    % Received signal
    Y = sqrt(snrLinear) * H * txMatrix + noise;

    %% ZF Detection
    W = pinv(H);

    detectedSymbols = W * Y;

    %% Scale back
    detectedSymbols = detectedSymbols / sqrt(snrLinear);

    %% Demodulation
    detectedBits = demodulateSignal( ...
        detectedSymbols(:), modulation);

    %% Match number of bits
    detectedBits = detectedBits(1:min(length(bits), ...
                  length(detectedBits)));

    originalBits = bits(1:length(detectedBits));

    %% Calculate BER
    BER(snrIndex) = ...
        sum(originalBits ~= detectedBits) / ...
        length(originalBits);

end

%% Plot BER
semilogy(SNR_dB, BER, '-o', ...
    'LineWidth', 1.5);

legendText{end+1} = ...
    sprintf('%dx%d MIMO', Nt, Nr);

fprintf('%s | %s | Correlation = %.1f | %dx%d MIMO\n', ...
    modulation, fading, rho, Nt, Nr);

fprintf('BER at 20 dB = %.5e\n', BER(end));
```

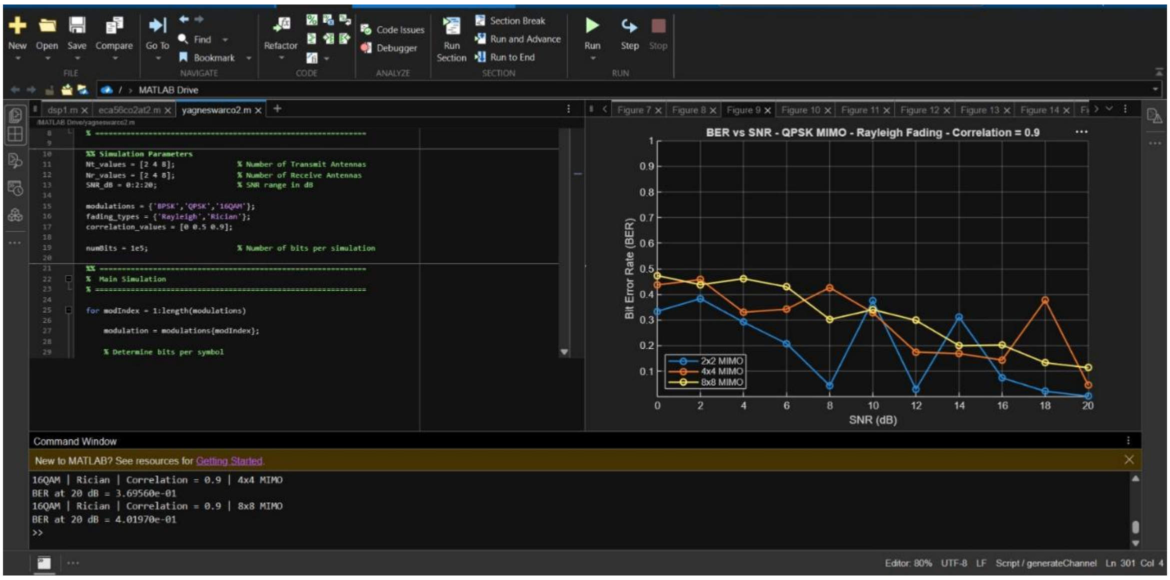
```
        end

        %% Plot Settings
        xlabel('SNR (dB)');
        ylabel('Bit Error Rate (BER)');

        title(sprintf( ...
            'BER vs SNR - %s MIMO - %s Fading - Correlation = %.1f', ...
            modulation, fading, rho));

        legend(legendText, 'Location', 'southwest');

        ylim([1e-5 1]);

        hold off;

    end
  end
end


%% ============================================================
% Function 1: Modulation
% ============================================================

function symbols = modulateSignal(bits, modulation)

  switch modulation

    case 'BPSK'

        symbols = 2*bits - 1;
        symbols = complex(symbols);

    case 'QPSK'

        % Make number of bits even
        if mod(length(bits),2) ~= 0
            bits = [bits; 0];
        end

        bitPairs = reshape(bits,2,[]).';

        I = 2*bitPairs(:,1)-1;
```

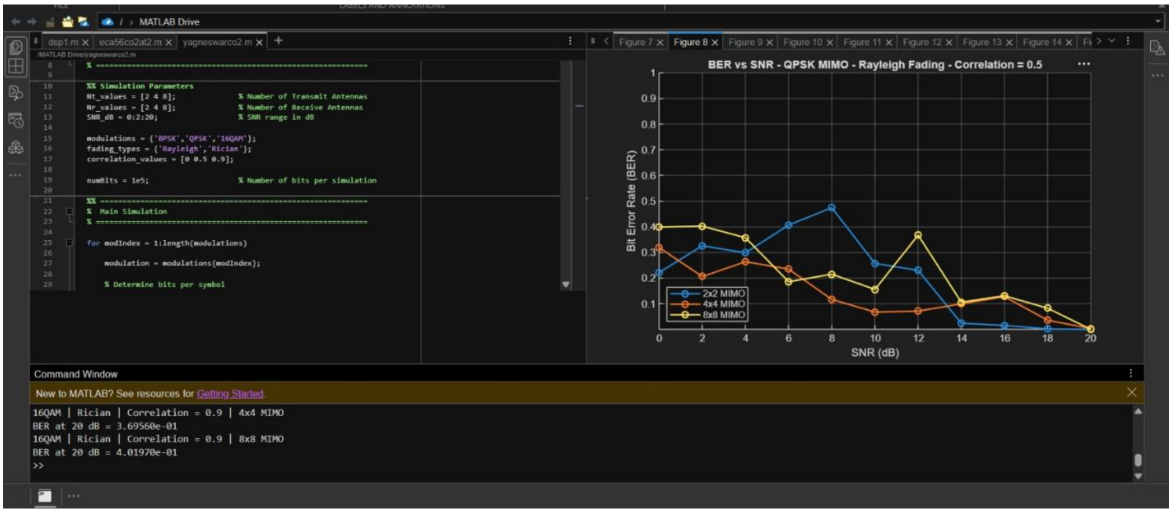
```
        Q = 2*bitPairs(:,2)-1;

        symbols = (I + 1i*Q)/sqrt(2);

      case '16QAM'

        % Make number of bits multiple of 4
        remainder = mod(length(bits),4);

        if remainder ~= 0
          bits = [bits; zeros(4-remainder,1)];
        end

        bitGroups = reshape(bits,4,[]).';

        % Gray coded 16-QAM mapping
        I = 2*(2*bitGroups(:,1)+bitGroups(:,2))-3;
        Q = 2*(2*bitGroups(:,3)+bitGroups(:,4))-3;

        symbols = (I + 1i*Q)/sqrt(10);

    end

    symbols = symbols(:);

end


%% ===================================================================
% Function 2: Demodulation
% ===================================================================

function bits = demodulateSignal(symbols, modulation)

    switch modulation

      case 'BPSK'

        bits = real(symbols) > 0;
        bits = bits(:);

      case 'QPSK'

        I = real(symbols) > 0;
```

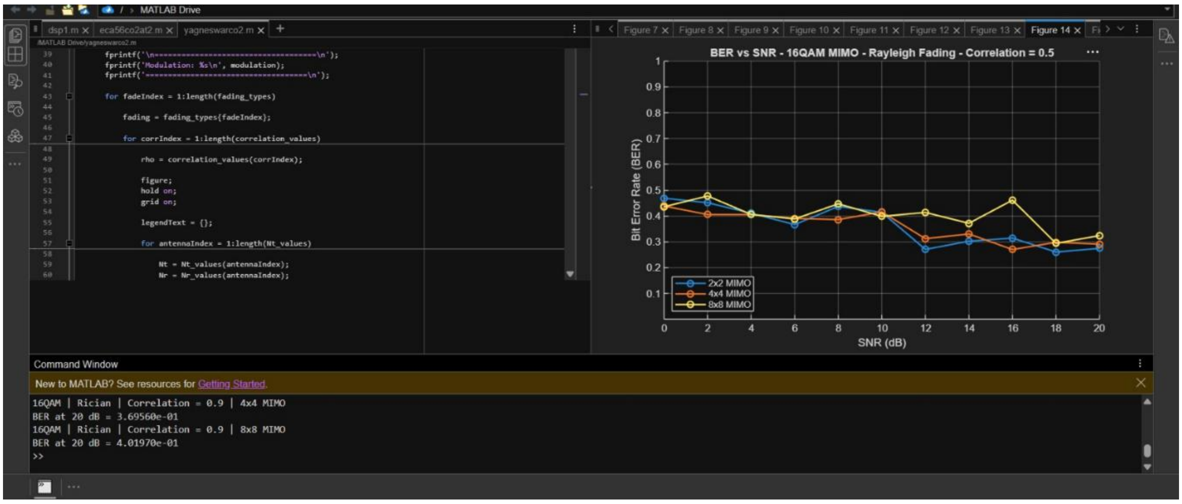
```
        Q = imag(symbols) > 0;

        bits = zeros(2*length(symbols),1);

        bits(1:2:end) = I;
        bits(2:2:end) = Q;

      case '16QAM'

        % Undo normalization
        symbols = symbols * sqrt(10);

        I = real(symbols);
        Q = imag(symbols);

        bits = zeros(4*length(symbols),1);

        % Decision levels
        I_bits = zeros(length(symbols),2);
        Q_bits = zeros(length(symbols),2);

        % I decision
        I_bits(I < -2,:) = repmat([0 0],sum(I < -2),1);
        I_bits(I >= -2 & I < 0,:) = repmat([0 1],sum(I >= -2 & I < 0),1);
        I_bits(I >= 0 & I < 2,:) = repmat([1 1],sum(I >= 0 & I < 2),1);
        I_bits(I >= 2,:) = repmat([1 0],sum(I >= 2),1);

        % Q decision
        Q_bits(Q < -2,:) = repmat([0 0],sum(Q < -2),1);
        Q_bits(Q >= -2 & Q < 0,:) = repmat([0 1],sum(Q >= -2 & Q < 0),1);
        Q_bits(Q >= 0 & Q < 2,:) = repmat([1 1],sum(Q >= 0 & Q < 2),1);
        Q_bits(Q >= 2,:) = repmat([1 0],sum(Q >= 2),1);

        bits(1:4:end) = I_bits(:,1);
        bits(2:4:end) = I_bits(:,2);
        bits(3:4:end) = Q_bits(:,1);
        bits(4:4:end) = Q_bits(:,2);

    end

    bits = double(bits(:));

end
```

```
%% ==============================================================
% Function 3: MIMO Channel Generation
% ==============================================================

function H = generateChannel(Nr, Nt, fading, rho)

    %% Independent Rayleigh channel
    H0 = (randn(Nr,Nt) + ...
          1i*randn(Nr,Nt))/sqrt(2);

    %% Apply transmit antenna correlation
    Rt = zeros(Nt,Nt);

    for i = 1:Nt
        for j = 1:Nt
            Rt(i,j) = rho^abs(i-j);
        end
    end

    %% Correlation matrix square root
    Rt_sqrt = sqrtm(Rt);

    %% Correlated channel
    H = H0 * Rt_sqrt;

    %% Rician Fading
    if strcmp(fading,'Rician')

        K = 5; % Rician K-factor

        H_LOS = ones(Nr,Nt);

        H = sqrt(K/(K+1))*H_LOS + ...
            sqrt(1/(K+1))*H;

    end

end
```

Output:


BER vs SNR - QPSK MIMO - Rayleigh Fading - Correlation = 0.9
Bit Error Rate (BER)
SNR (dB)
2x2 MIMO
4x4 MIMO
8x8 MIMO
Command Window
16QAM | Rician | Correlation = 0.9 | 4x4 MIMO
BER at 20 dB = 3.69560e-01
16QAM | Rician | Correlation = 0.9 | 8x8 MIMO
BER at 20 dB = 4.01970e-01


BER vs SNR - QPSK MIMO - Rayleigh Fading - Correlation = 0.5
Bit Error Rate (BER)
SNR (dB)
2x2 MIMO
4x4 MIMO
8x8 MIMO
Command Window
16QAM | Rician | Correlation = 0.9 | 4x4 MIMO
BER at 20 dB = 3.69560e-01
16QAM | Rician | Correlation = 0.9 | 8x8 MIMO
BER at 20 dB = 4.01970e-01


BER vs SNR - 16QAM MIMO - Rayleigh Fading - Correlation = 0.5
Bit Error Rate (BER)
SNR (dB)
2x2 MIMO
4x4 MIMO
8x8 MIMO
Command Window
16QAM | Rician | Correlation = 0.9 | 4x4 MIMO
BER at 20 dB = 3.69560e-01
16QAM | Rician | Correlation = 0.9 | 8x8 MIMO
BER at 20 dB = 4.01970e-01